

Lecture 1

BNFO 135

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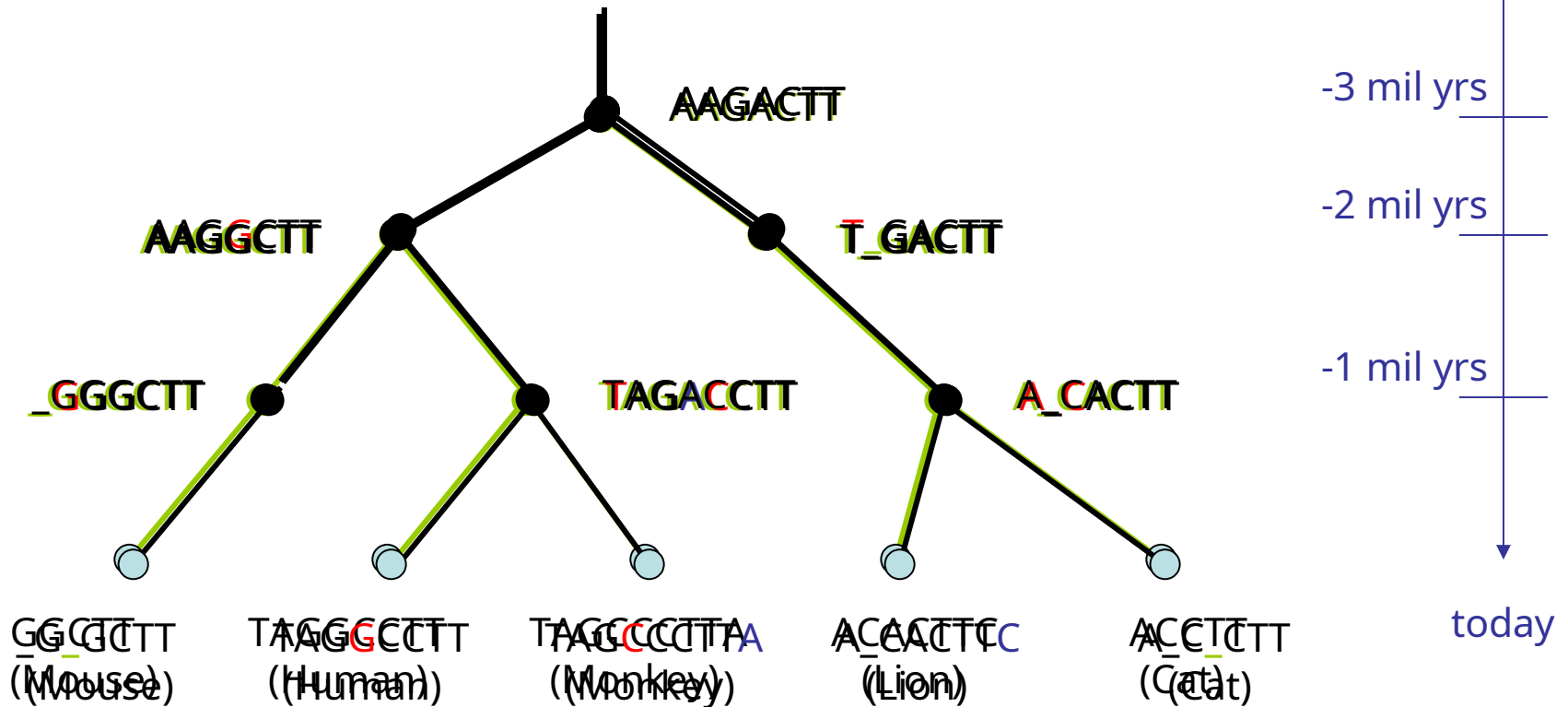
Course overview

- Python programming language (and some Unix basics)
 - Unix basics
 - Intro Python exercises
 - Programs for comparing DNA and protein sequences
- Sequence analysis
 - Pairwise and multiple sequence comparison
 - Sequence alignments
 - Application of alignments
 - Heuristic alignment (BLAST)

Overview (contd)

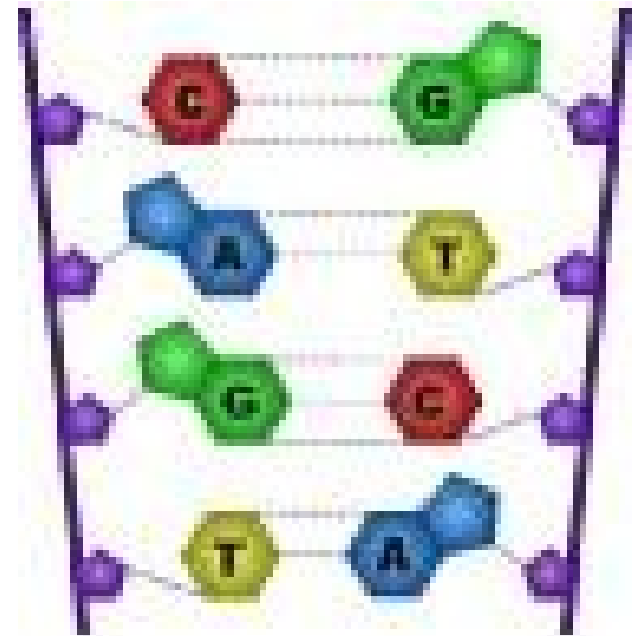
- Grade: 15% programming assignments, two 25% mid-terms and 25% final exam
- Recommended Texts:
 - Bioinformatics Programming using Python by Mitchell L Model

Nothing in biology makes sense, except in the light of evolution

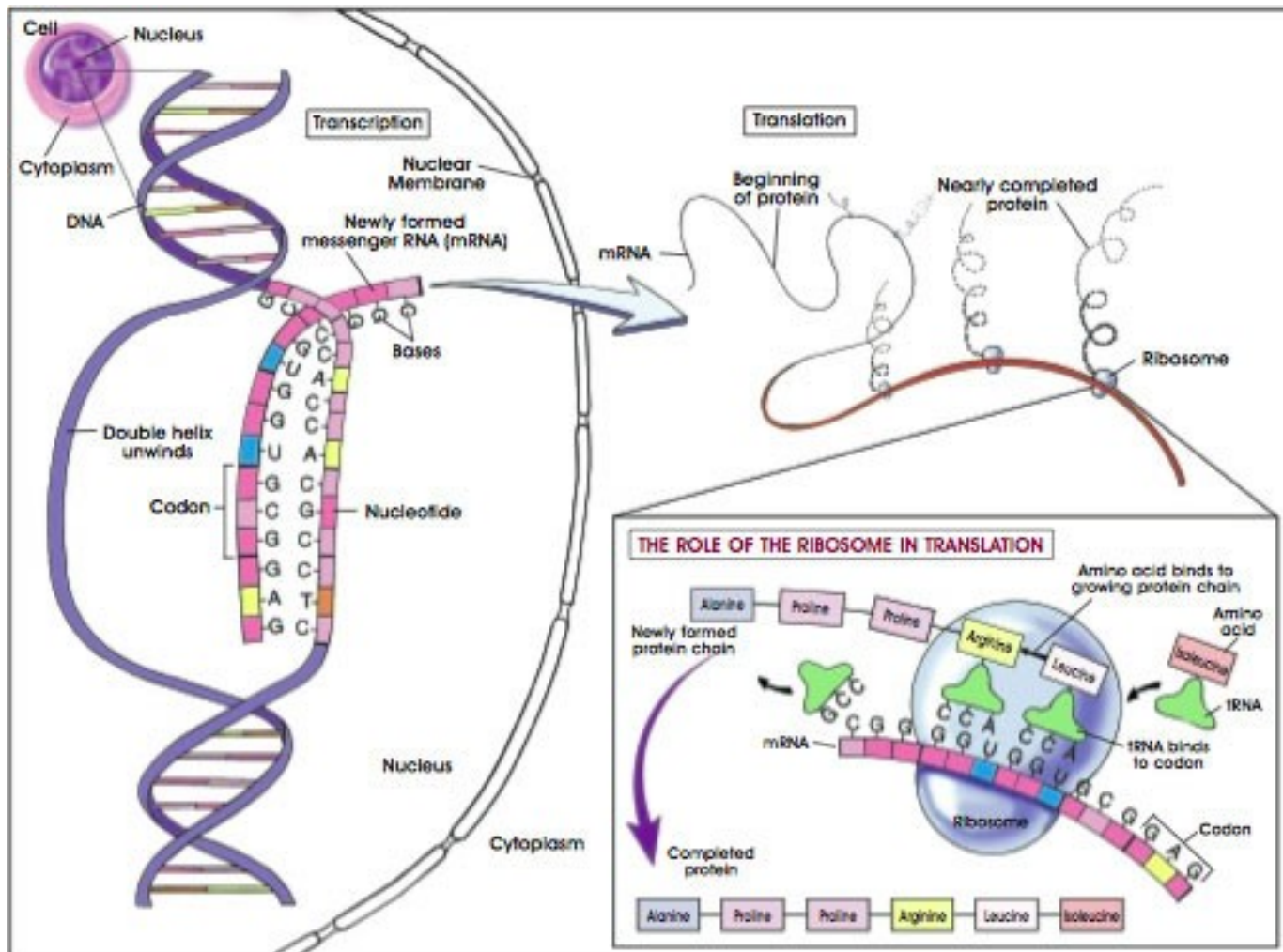


Representing DNA in a format manipulatable by computers

- DNA is a double-helix molecule made up of four nucleotides:
 - Adenosine (A)
 - Cytosine (C)
 - Thymine (T)
 - Guanine (G)
- Since A (adenosine) always pairs with T (thymine) and C (cytosine) always pairs with G (guanine) knowing only one side of the ladder is enough
- We represent DNA as a sequence of letters where each letter could be A, C, G, or T.
- For example, for the helix shown here we would represent this as CAGT.



Transcription and translation



Amino acids

Proteins are chains of amino acids. There are twenty different amino acids that chain in different ways to form different proteins.

For example,
FLLVALCCRFGH
(this is how we could store it in a file)

This sequence of amino acids folds to form a 3-D structure

	T	C	A	G
T	TTT Phe (F) TTC " TTA Leu (L) TTG "	TCT Ser (S) TCC " TCA " TCG "	TAT Tyr (Y) TAC " TAA Ter TAG Ter	TGT Cys (C) TGC " TGA Ter TGG Trp (W)
C	CTT Leu (L) CTC " CTA " CTG "	CCT Pro (P) CCC " CCA " CCG "	CAT His (H) CAC " CAA Gln (Q) CAG "	CGT Arg (R) CGC " CGA " CGG "
A	ATT Ile (I) ATC " ATA " ATG Met (M)	ACT Thr (T) ACC " ACA " ACG "	AAT Asn (N) AAC " AAA Lys (K) AAG "	AGT Ser (S) AGC " AGA Arg (R) AGG "
G	GTT Val (V) GTC " GTA " GTG "	GCT Ala (A) GCC " GCA " GCG "	GAT Asp (D) GAC " GAA Glu (E) GAG "	GGT Gly (G) GGC " GGA " GGG "

Protein folding

DNA

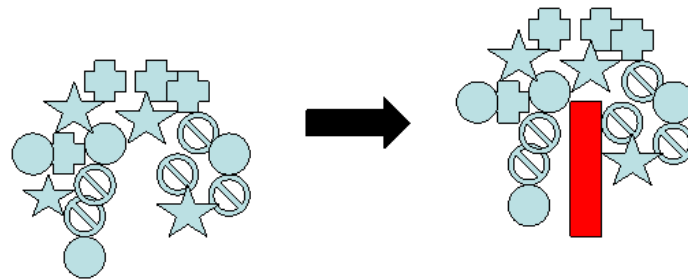
ATGTGCGAATTCCGGTTACGGTTGTGAGC



Amino acid chain

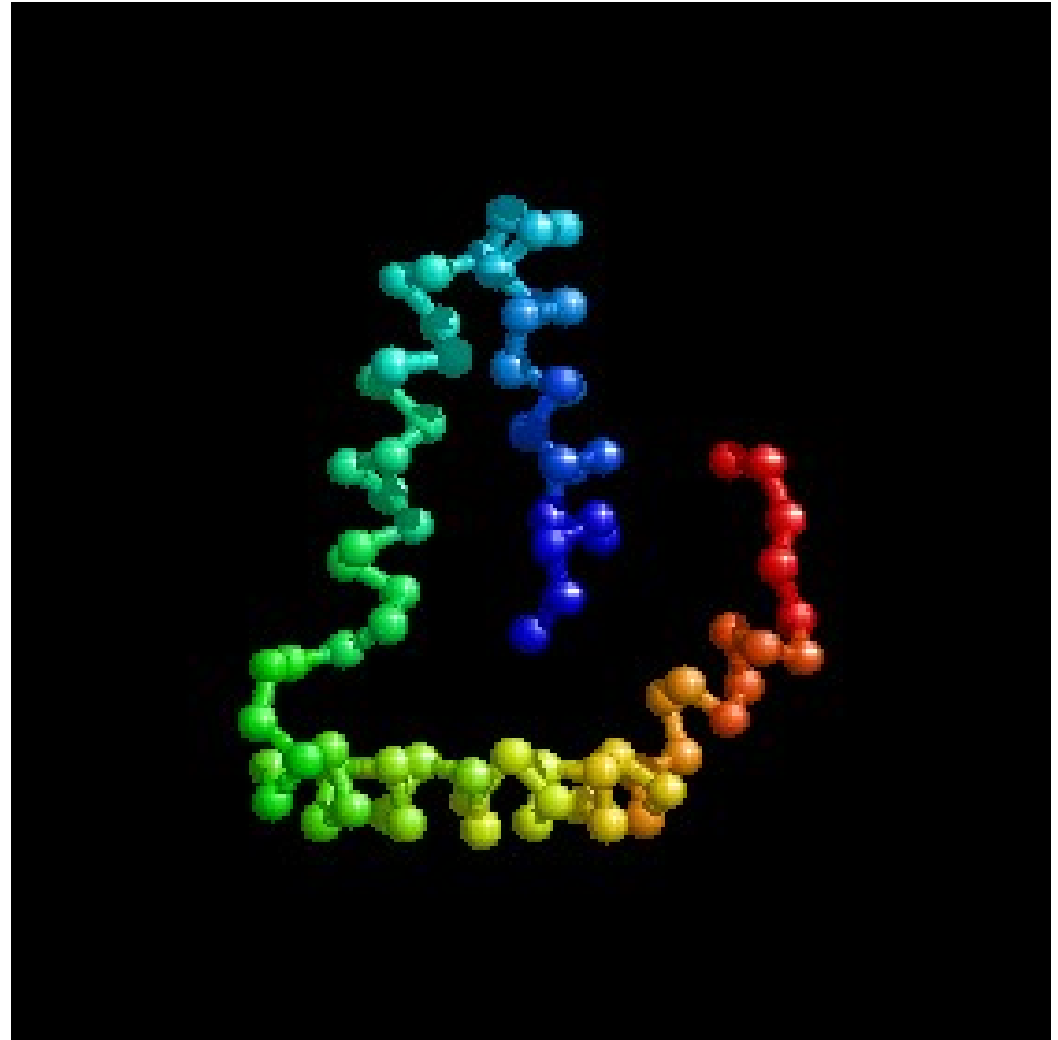


Folded protein with
binding site



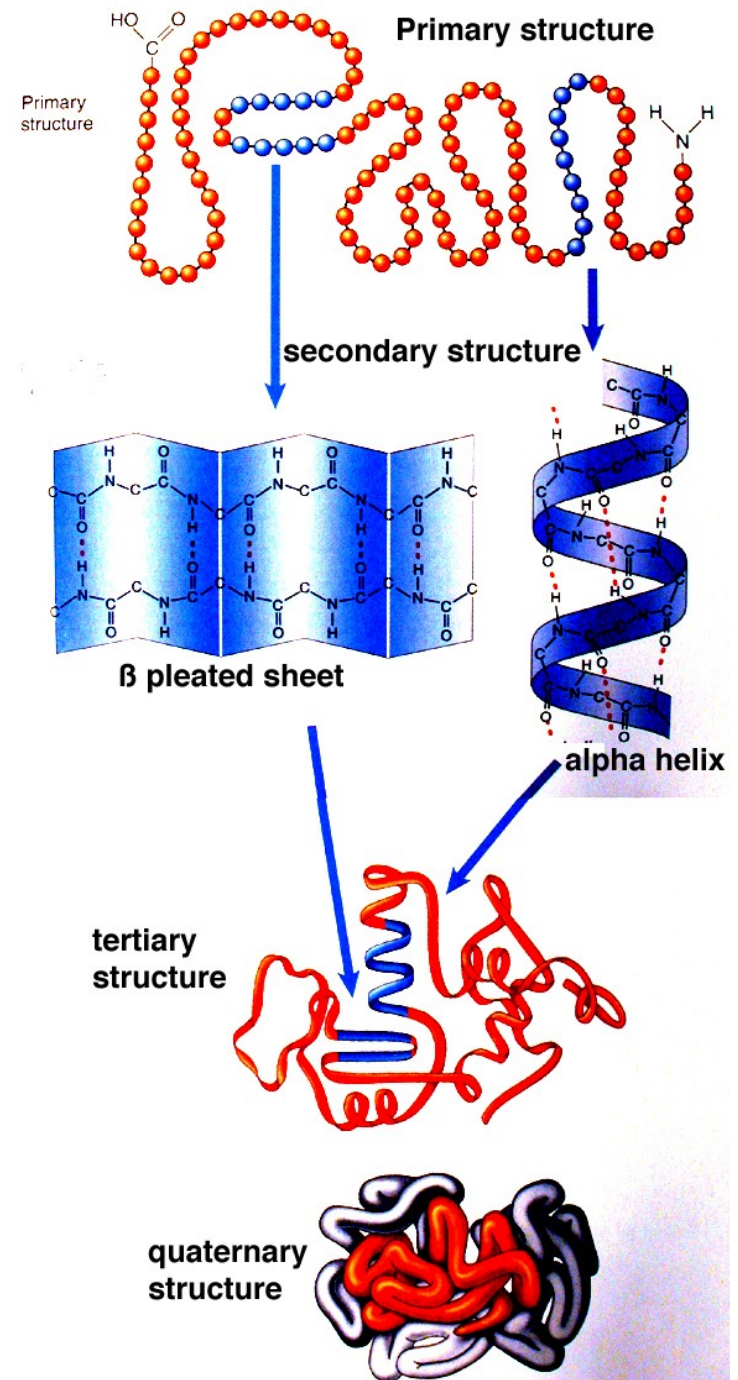
Protein folding

- The protein folding problem is to determine the 3-D protein structure from the sequence.
- Experimental techniques are very expensive.
- Computational are cheap but difficult to solve.
- By comparing sequences we can deduce the evolutionary conserved portions which are also functional (most of the time).



Protein structure

- Primary structure: sequence of amino acids.
- Secondary structure: parts of the chain organizes itself into alpha helices, beta sheets, and coils. Helices and sheets are usually evolutionarily conserved and can aid sequence alignment.
- Tertiary structure: 3-D structure of entire chain
- Quaternary structure: Complex of several chains



Key points

- DNA can be represented as strings consisting of four letters: A, C, G, and T. They could be very long, e.g. thousands and even millions of letters
- Proteins are also represented as strings of 20 letters (each letter is an amino acid). Their 3-D structure determines the function to a large extent.